

LANGUAGE LEARNING APP

Introduction

The Language Learning App is an interactive educational application designed to help users learn new languages easily and effectively. The application provides vocabulary lessons, flashcards, pronunciation support, quizzes, and progress tracking.

The main goal of the project is to make language learning simple, engaging, and user-friendly. Users can practice words, phrases, and grammar through daily lessons and quizzes. The application stores learning progress locally using browser local storage and can also be connected to Firebase for online data synchronization.

Objective

The objectives of the project are:

- To help users learn new languages interactively.
- To improve vocabulary and pronunciation skills.
- To provide quizzes for self-assessment.
- To track user learning progress.
- To create a clean and responsive user interface.
- To store user data locally or through Firebase backend.

Features of the Application

1. Flashcards
 - Displays new words with translations.
 - Includes pronunciation support using speech synthesis.
 - Helps users memorize vocabulary quickly.
2. Daily Lessons
 - Vocabulary lessons
 - Grammar lessons
 - Phrase practice
 - Pronunciation exercises

3. Quiz Section

- Multiple-choice questions
- Instant answer checking
- Score calculation

4. Progress Tracking

- Displays completed lessons
- Shows quiz score
- Progress bar visualization

5. Local Storage Support

- Saves quiz score and learning data.
- Data remains available after page refresh.

Technologies Used

1. HTML (HyperText Markup Language)

HTML is used to create the structure of the Language Learning App. It helps in designing pages such as lessons, flashcards, quizzes, and progress sections.

2. CSS (Cascading Style Sheets)

CSS is used for styling the application. It improves the appearance of the app by adding colors, layouts, animations, responsive design, and attractive user interfaces.

3. JavaScript

JavaScript is used to add functionality and interactivity to the application. It handles:

- Flashcard interactions
- Quiz logic
- Score calculation
- Pronunciation feature
- Dynamic content updates

4. Local Storage

Local Storage is used to save user learning progress, quiz scores, and completed lessons directly in the browser. It allows data to remain available even after refreshing the page.

5. Speech Synthesis API

The Speech Synthesis API is used for pronunciation support. It converts text into speech so users can hear correct pronunciation of words and phrases.

Source code

```
<!DOCTYPE html>

<html lang=" en" >

<head>

<meta charset=" UTF-8" >

<meta name=" viewport" content=" width=device-width,initial-scale=1.0" >

<title>Language Learning App</title>


<style>

  Body{

    Font-family: Arial, sans-serif;

    Background: #f4f4f4;

    Text-align: center;

    Padding: 20px;

  }


  .container{

    Background: white;

    Width: 350px;

    Margin: auto;

    Padding: 20px;

    Border-radius: 10px;

    Box-shadow: 0 0 10px gray;
```



```
}
```

```
H1{
```

```
    Color:#333;
```

```
}
```

```
.card{
```

```
    Margin:20px 0;
```

```
    Padding:20px;
```

```
    Background:#e3f2fd;
```

```
    Border-radius:10px;
```

```
}
```

```
Button{
```

```
    Padding:10px 15px;
```

```
    Margin:5px;
```

```
    Border:none;
```

```
    Background:#2196f3;
```

```
    Color:white;
```

```
    Border-radius:5px;
```

```
    Cursor:pointer;
```

```
}
```

```
Button:hover{
```

```
    Background:#0b7dda;
```

```
}
```

```
#score{
    Font-size:20px;
    Color:green;
}
</style>
</head>

<body>

<div class=" container" >
    <h1>Language Learning App</h1>

    <div class=" card" >
        <h2 id=" word" >Hello</h2>
        <p id=" meaning" >Vanakkam</p>
    </div>

    <button onclick=" nextWord()" >Next Word</button>
    <button onclick=" speakWord()" >Pronounce</button>

    <hr>

    <h2>Quiz</h2>

    <p>What is the meaning of "Thank You" ?</p>
```

```
<button onclick=" checkAnswer( 'Nandri' )" >Nandri</button>
```

```
<button onclick=" checkAnswer( 'Bye' )" >Bye</button>
```

```
<h3 id=" result" ></h3>
```

```
<h3 id=" score" >Score:0</h3>
```

```
</div>
```

```
<script>
```

```
Const words =[
```

```
    {english:" Hello" ,tamil:" Vanakkam" },
```

```
    {english:" Thank You" ,tamil:" Nandri" },
```

```
    {english:" Good Morning" ,tamil:" KaalaiVanakkam" }
```

```
];
```

```
Let index =0;
```

```
Let score =localStorage.getItem( "score" )||0;
```

```
Document.getElementById( "score" ).innerHTML= "Score: "+ score;
```

```
Function nextWord(){
```

```
    Index =(index + 1) % words.length;
```

```
Document.getElementById( "word" ).innerHTML=
```

```
    Words[index].english;
```



```
Document.getElementById( "meaning" ).innerHTML =  
    Words[index].tamil;  
}
```

```
Function speakWord(){  
    Let text = document.getElementById( "word" ).innerHTML;  
  
    Let speech = new SpeechSynthesisUtterance(text);  
  
    speechSynthesis.speak(speech);  
}
```

```
Function checkAnswer(answer){  
  
    If(answer=== "Nandri" ){  
        Document.getElementById( "result" ).innerHTML =  
            "Correct Answer!" ;  
  
        Score++;  
    }  
    Else{  
        Document.getElementById( "result" ).innerHTML =  
            "Wrong Answer!" ;  
    }  
}
```

```
localStorage.setItem( "score" ,score);
```

```
document.getElementById( "score" ).innerHTML =  
    "Score: " + score;  
}  
  
</script>  
  
</body>  
</html>
```

Working of the Application

- User opens the application.
- User selects lessons or flashcards.
- The application displays words and translations.
- User listens to pronunciation using speech synthesis.
- User takes quizzes to test knowledge.
- Quiz scores and progress are saved locally.
- User can continue learning anytime

Advantages

- Easy to use
- Interactive learning experience
- Improves vocabulary and pronunciation
- Instant quiz feedback
- Responsive design for mobile and desktop
- Offline support using local storage

Disadvantages

Limited vocabulary in basic version

Requires internet for Firebase synchronization

No advanced AI translation features

Pronunciation depends on browser support